Stormwater Regulation Summary: Seattle, WA

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary driver: Salmon habitat protection and Puget Sound water quality.
- Seattle Stormwater Code (SMC 22.800) + King County Surface Water Design Manual (SWDM).
- Applies to ≥2,000 SF new + replaced impervious surface — one of the lowest thresholds in the US.
- Low Impact Development (LID) is required as the first priority — must demonstrate infeasibility before using conventional BMPs.
- Flow Duration Standard is the most stringent flow control requirement: match forested condition for ½ of 2-yr through 50-yr peaks.

2. Typical Soil Conditions

- Glacial deposits dominate — Vashon till (dense, compacted) overlying Vashon advance outwash (permeable gravels).
- Typical soils: Alderwood gravelly sandy loam (till over outwash), Everett gravelly sandy loam (outwash).
- Hydrologic Soil Group: C-D for Alderwood (till); A-B for Everett and similar outwash soils.
- Seattle has highly variable soil conditions — till is nearly impermeable while outwash gravels drain rapidly.
- Implication: Infiltration is excellent where outwash is accessible but impractical on till. Many urban sites have been regraded, mixing soil types.
- *Geotechnical borings critical — till depth varies significantly, sometimes within a single site.*

3. Green Roof Requirements

- Green roofs are not mandatory but qualify as LID BMPs — satisfying Seattle's first-priority LID requirement.
- RainWise rebate program: up to ~$7.90/SF for green roofs in eligible combined sewer basins (rates increased significantly in 2023-2024).
- Seattle Living Building Ordinance encourages green roofs in certain zones.
- SPU credits green roofs for both retention (evapotranspiration) and flow control (detention).
- WA State Clean Energy Transformation Act (CETA) and net metering support bio-solar systems.
- On constrained urban sites, green roofs may be the only feasible LID option — strengthens the case.

4. TSS Requirements

- Seattle requires Enhanced Treatment (for basic water quality) or equivalent: 80% TSS removal standard.
- Pollution-generating impervious surfaces (PGIS) — roads, parking, industrial — require treatment.
- Green roofs are credited for TSS removal on non-pollution-generating surfaces.
- For PGIS, additional treatment (bioretention, media filter) is typically required beyond what a green roof provides.
- *Seattle distinguishes between basic, enhanced, and oil-control treatment levels — green roof credit level should be confirmed in current SPU guidance.*

5. Nature-Based Retention Requirements

- LID Performance Standard: Infiltrate, disperse, or retain stormwater on-site to match pre-development (forested) hydrology.
- Flow Duration Standard: Post-development flow duration must match forested condition for range from 50% of 2-yr to 50-yr peak.
- This is one of the most stringent standards in the US — equivalent to retaining the vast majority of annual rainfall volume.
- Seattle uses continuous simulation modeling (WWHM or MGS Flood) rather than single-event CN method.
- CN is not the primary design tool — Western Washington Hydrology Model (WWHM) runs continuous hourly precipitation for 50+ years of record.
- *Design methodology is significantly different from East Coast/Midwest CN-based approach — ensure design team uses WWHM.*

6. Allowable Outflow Rates

- Flow Duration Standard: Outflow at any rate between 50% of the 2-yr peak and the 50-yr peak must not exceed the duration of that flow rate under forested conditions.
- This is NOT a simple peak rate limit — it controls the entire flow-duration curve.
- For simplified reference: Matching forested condition typically limits effective release to approximately 5-15 l/s/ha for the 2-year storm.
- *The flow duration standard cannot be accurately expressed as a single l/s/ha value — continuous simulation is required.*
- King County Surface Water Design Manual provides detailed methodology and flow control targets.